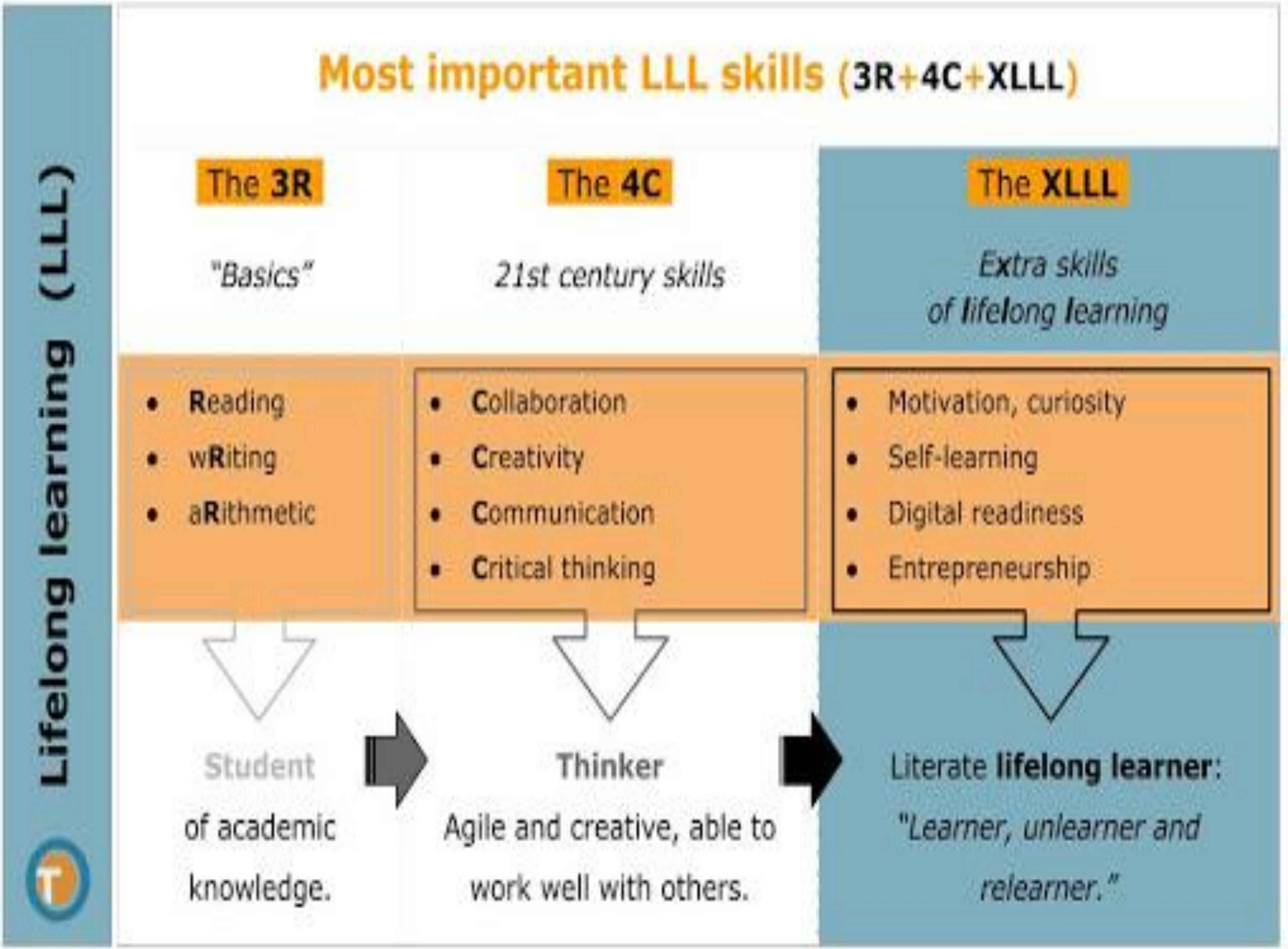
The heart of All Topics is the Learning Activities

اعداد

د.نرمين الدفراوي

أستاذ المناهج وطرق تدريس العلوم المساعد


Lifelong learning (LLL)
Most important LLL skills (3R+4C+XLLL)
The 3R
"Basics"
Reading
wRiting
aRithmetic
Student
of academic knowledge.
The 4C
21st century skills
Collaboration
Creativity
Communication
Critical thinking
Thinker
Agile and creative, able to work well with others.
The XLLL
Extra skills of lifelong learning
Motivation, curiosity
Self-learning
Digital readiness
Entrepreneurship
Literate lifelong learner:
"Learner, unlearner and relearner."

~~Active Learning~~

Interactive Teaching
and
Effective Learning

Interactive Teaching

- Telling is not teaching, nor is listening learning.
- You must engage students in learning activities that lead to a higher level of understanding and result in the participant's.
- ability to apply what students learned on the job
- Interactive teaching is a two-way process of active participant (students) engagement with each other, the facilitator(teacher), and the content(curriculum).

Why Interactive teaching?

- "Tell me, I forget. Show me, I remember. Involve me, I understand." Tse Lao *Chinese Proverb*

حدثني سأنسي، أرني سأتذكر أشركني سأفهم

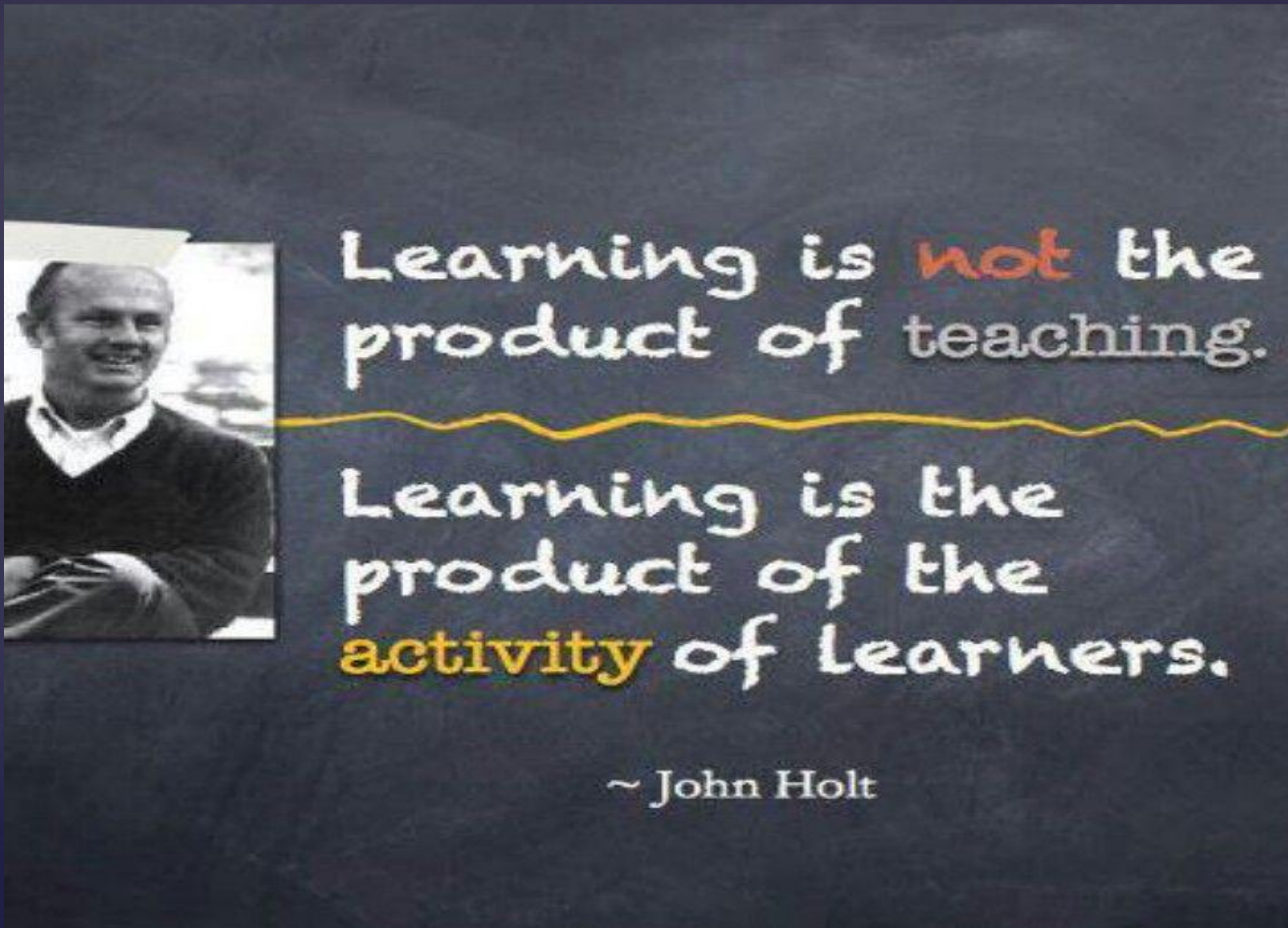
"What children can **do together** today they can do alone tomorrow." *Vygotsky, 1965*

ما يستطيع أن يفعله الأطفال مع بعض اليوم، سيقومون به لوحدهم بعد ذلك

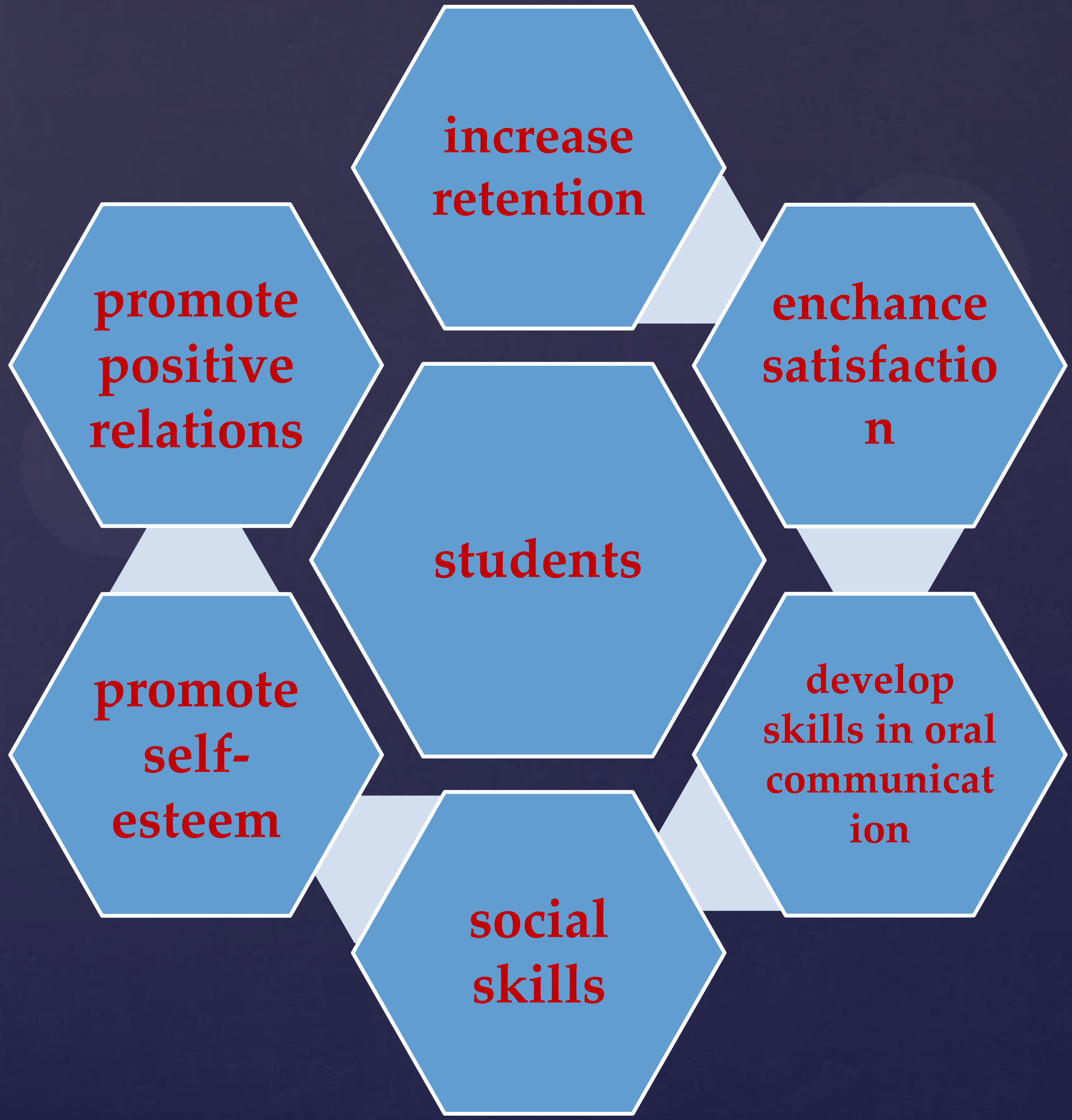
"What is the most effective method of teaching? Students teaching other students."

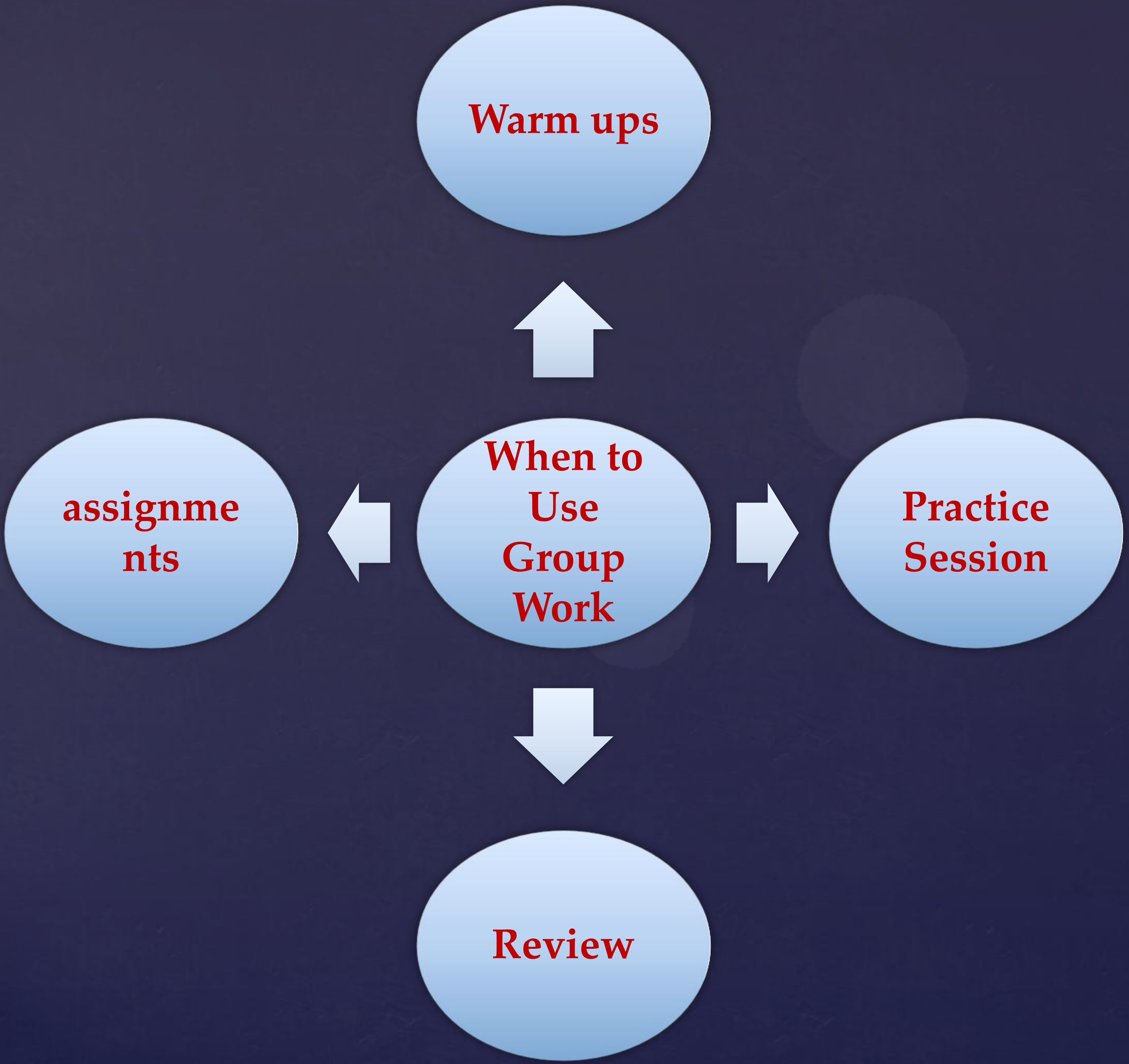
التعلم بالأقران

McKeachie, Pintrich, Lin, & Smith, 1986


Learning is not the product of teaching.
Learning is the product of the activity of learners.
~ John Holt

Cooperative Learning




Warm ups
When to Use Group Work
assignments
Practice Session
Review

Where to Start…

- Start with clear learning outcomes

Helps you plan session and helps participants by providing clear view of the session’s direction

Specific	Measurable	Achievable	Realistic	Timely
S	**M**	**A**	**R**	**T**
G	O	A	L	S
What do you want to do?	How will you know when you've reached it?	Is it in your power to accomplish it?	Can you realistically achieve it?	When exactly do you want to accomplish it?

Learning Outcomes

- Think of the "ideal" students or graduates

  - What students know?>> K
  - What students can do? ?>> P
  - What students care about ? ?>> A

Knowledge, Attitude, and Practice

Questions for Instructional Design

1. **What do you want the student to be able to do? (learning Outcomes)**

2- What activity will facilitate the learning? (Pedagogy—learning & teaching)

3- How will the *student demonstrate* the learning? (Assessment)

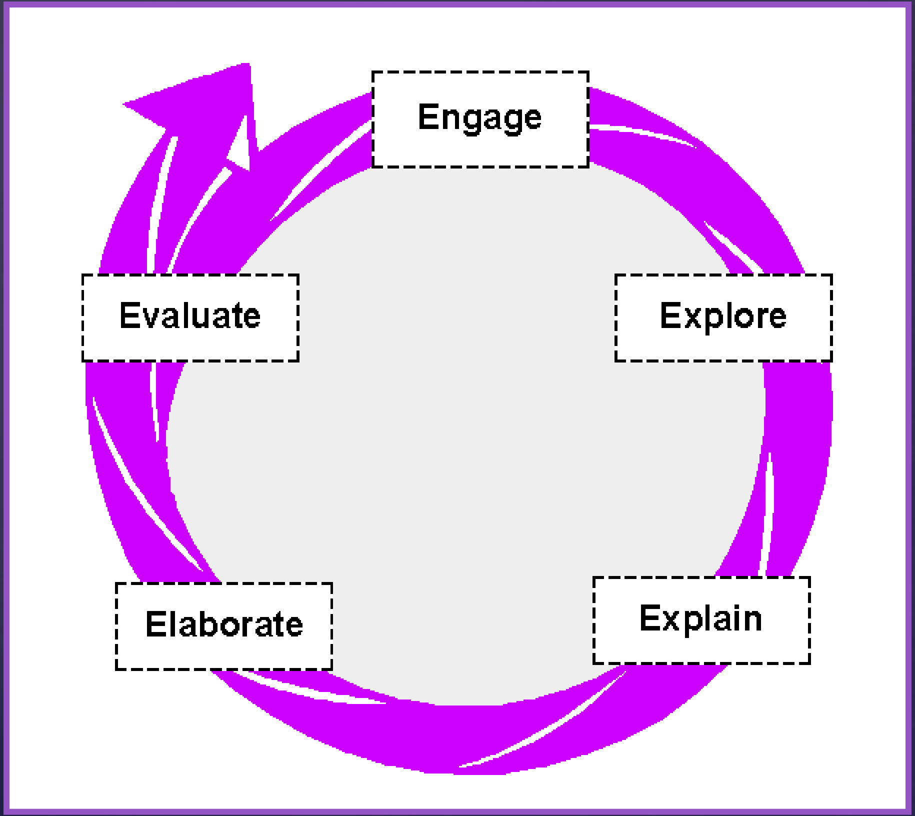
4- How will the *teacher* know the student has done this well? (Criteria)

How do I link
Learning Outcomes
to
Teaching Methods
and
Learning Activities

Teaching Methods

Brainstorming	• possible problems • possible solutions
Problem Solving Method	• critical thinking • taking decision • reflective thinking.
5 E' S	• Doing as scientist do.

Learning Cycle



Engage student interest.

Explore content and learn lab skills through relevant and concrete experiences

Explain questions generated by introducing content

Elaborate by applying concepts and lab skills to new inquiry situations

Evaluate content, process, and communication skills

Brainstorming Technique

- **Free writing**

When you have

NO ideas / TOO MANY ideas

about a topic.

Options:

Write for a specific **time** period

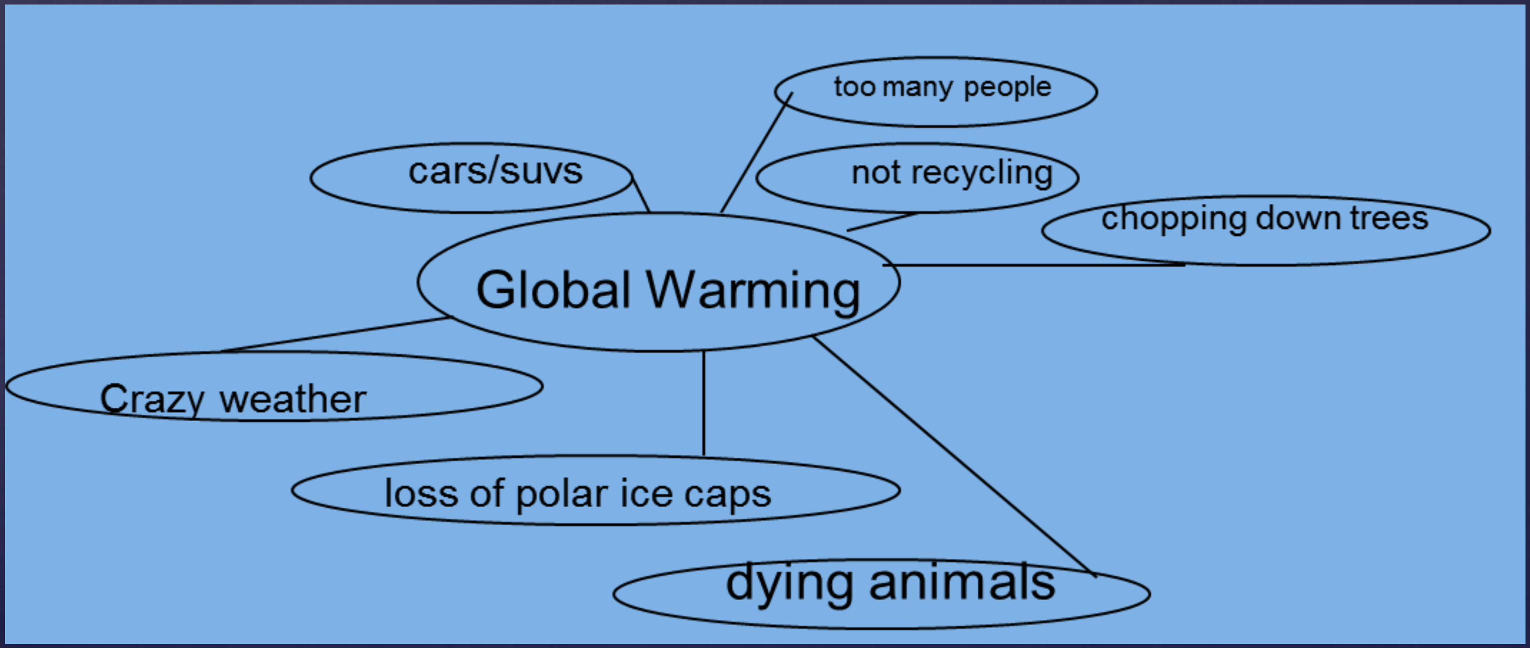
- **T-Diagram**

This technique helps when you are writing a **contrast** or **comparison** essay.

Brainstorming Technique

➢ Spoke Diagram

Use this technique when you want to explore *cause and effect*.



Problem Solving Techniques

- **Induction:**

It is like teaching with **discovering** method.

The similarities and dissimilarities are found.

Then you can reach the general **rule** or **law** with the techniques "generalization"

- **Deduction:**

Some *general laws* and **rules** which are reached before are given to the students and want them *to apply this method to different singular*

Interactive Teaching Techniques

- Think/Pair/Share
- Incident Process
- Muddiest Point
- Short writing exercises
- Note Review
- Demonstration
- KWL.
- One-Minute Papers
- mysterious

-Three-step interview
- Misconception Check
-.Three-minute review
-.Round robin
-.Numbered heads
-.Circle the sage
- Picture Prompt

1- One-Minute Papers:

Students write for one minute on a specific question (which might be generalized to

"what was the most important thing you learned today".

Best used at the end of the class session.

2- Muddiest Point:

Like the Minute Paper, but asks for the "**most confusing**" point instead.

Best used at the end of main idea/or the class session..

3- Misconception Check:

Discover class's preconceptions.

Useful for **starting** new subject.

4- Drawing for Understanding:

Students **illustrate an abstract concept** or idea.

Comparing **drawings** around the room **can clear up misconceptions**.

5-Circle the Questions:

Pre-make a handout that has a few dozen likely student **questions** (make them specific) **on your topic** for that day and ask students to **circle the ones they don't know the answers** to, then turn in the paper.

6-Ask the Winner:

Ask **students** to silently **solve a problem on the board**.
After revealing the answer, instruct those **who got it right to raise their hands** (and keep them raised); then, all **other students are to talk** to someone with a **raised hand to better understand** the question and how to solve it next time..

7- True or False? :

Distribute cards (one to each student) on which is written a statement.

Half of the cards will contain **statements** that are **true**, **half false**.

Students decide if theirs is one of the true statements or not, using whatever means they desire.

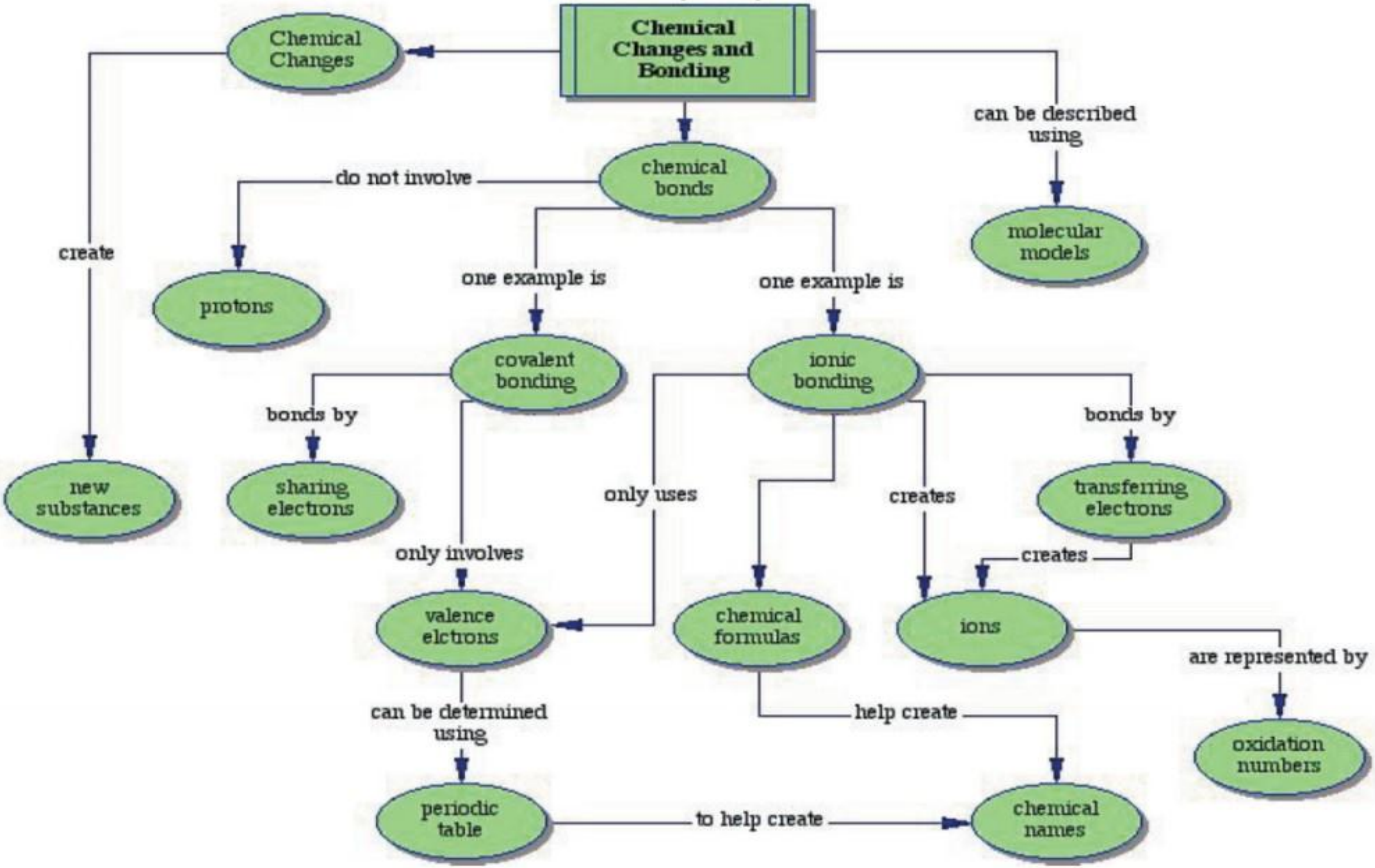
* designate half the room a space for those who think their statements are true, and the other half for false..

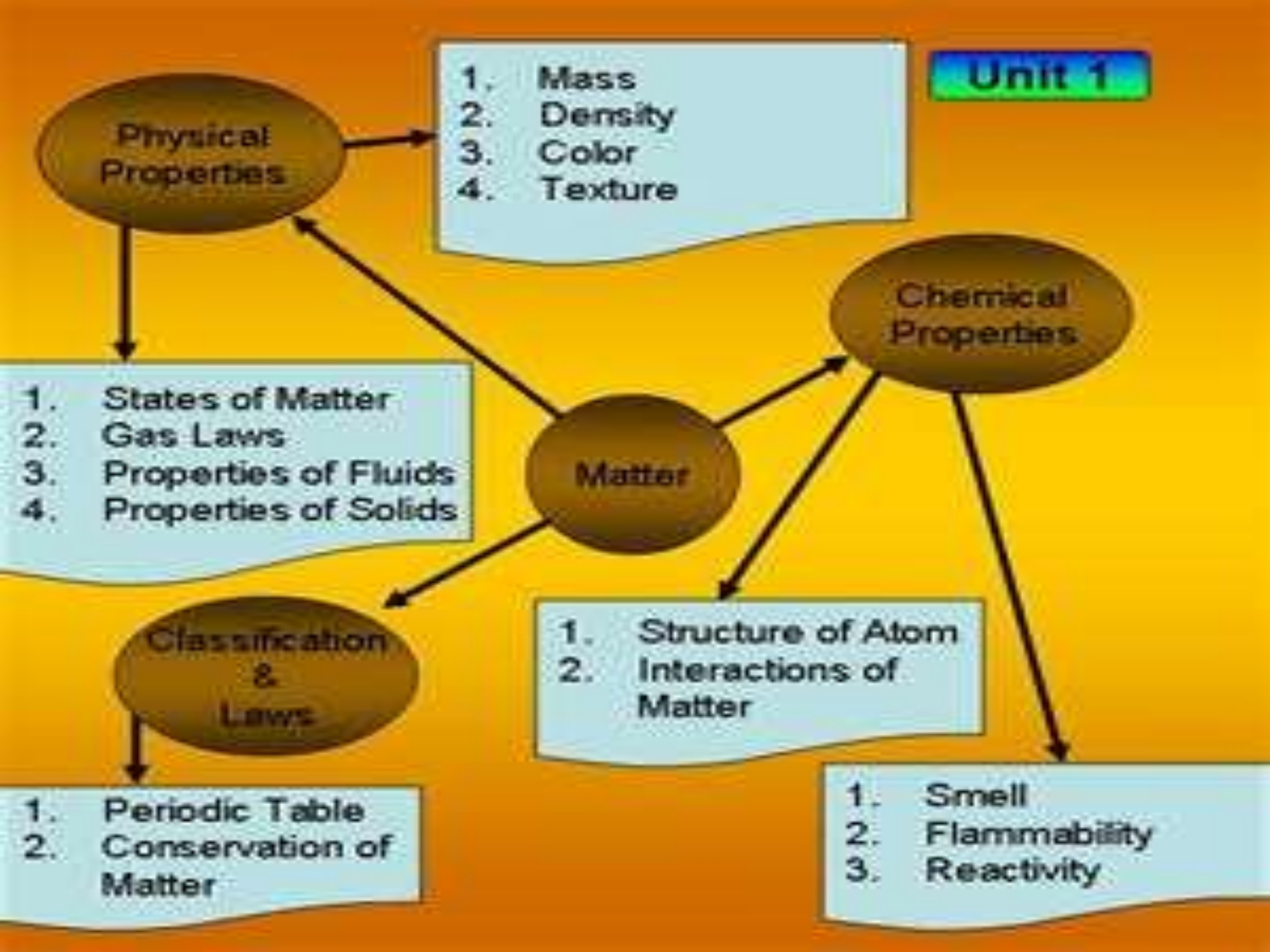
8- Concept Mapping :

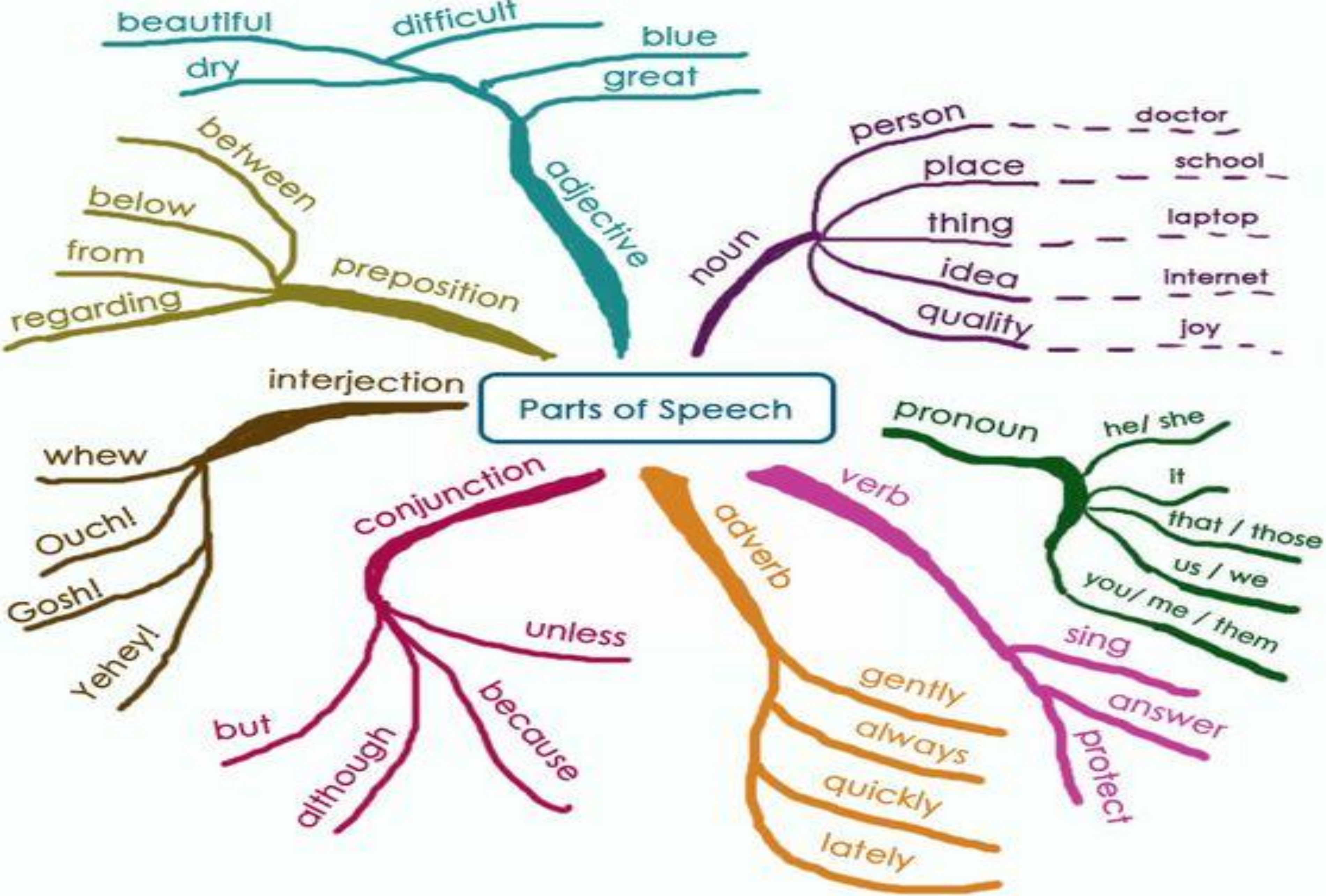
Students write **keywords** onto sticky notes and then **organize them** into a flowchart.

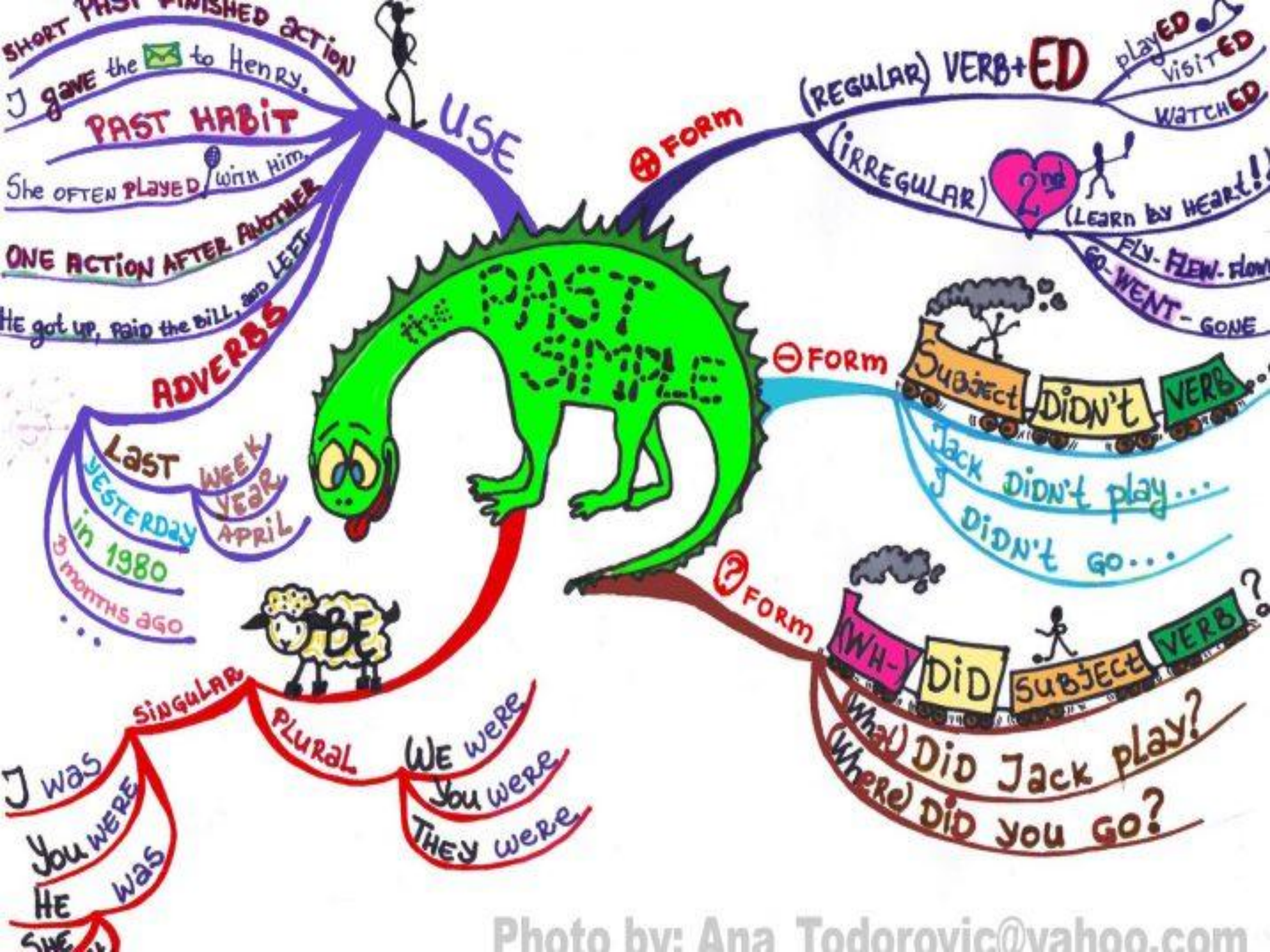
Could be less structured: students simply draw the connections they make between concepts..

Chemical Changes and Bonding Concept Map

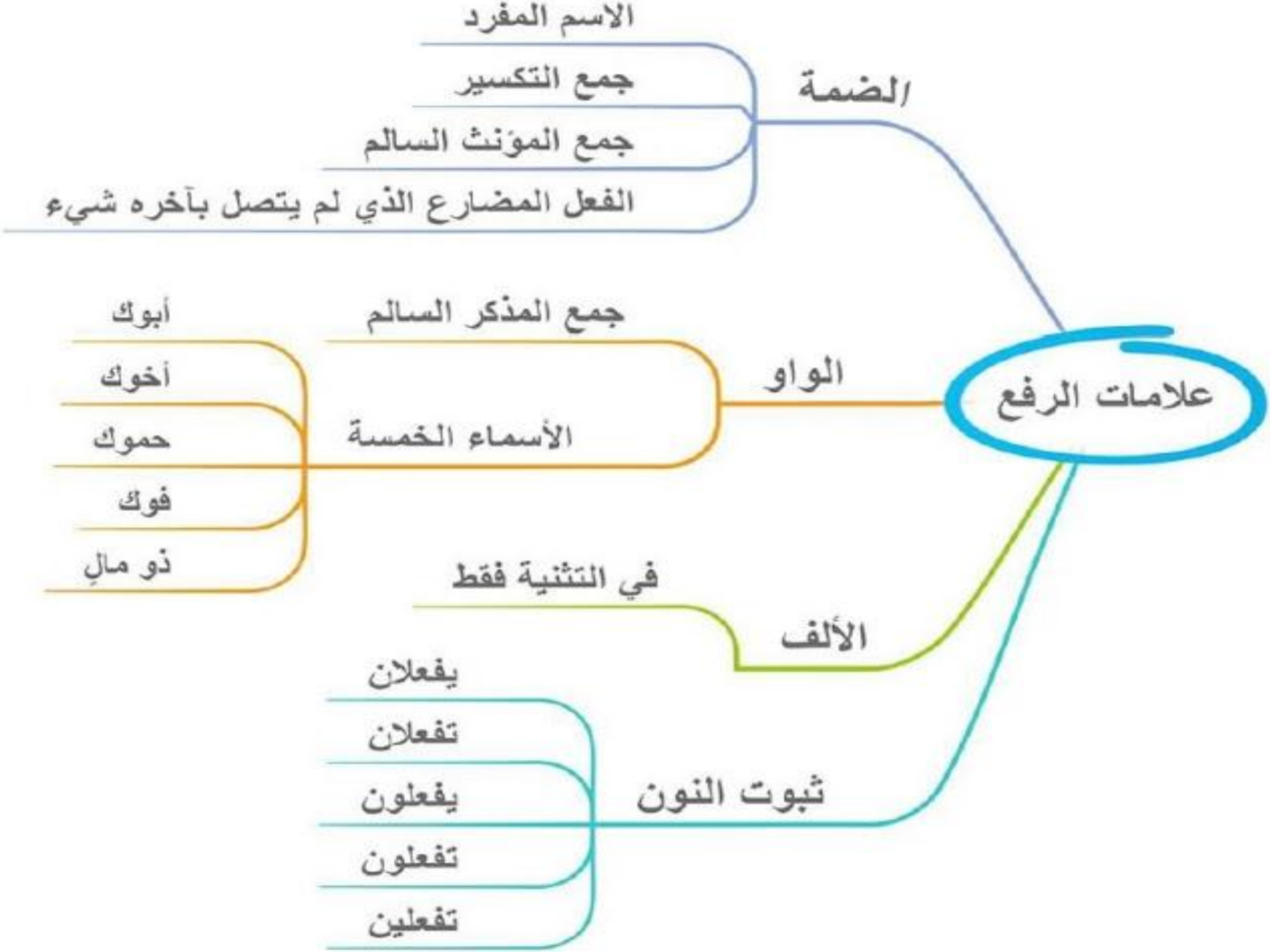



Unit 1
Physical Properties
1. Mass
2. Density
3. Color
4. Texture
Matter
Chemical Properties
1. States of Matter
2. Gas Laws
3. Properties of Fluids
4. Properties of Solids
Classification & Laws
1. Periodic Table
2. Conservation of Matter
1. Structure of Atom
2. Interactions of Matter
1. Smell
2. Flammability
3. Reactivity


Parts of Speech
adjective
beautiful
difficult
blue
dry
great
noun
person
place
thing
idea
quality
doctor
school
laptop
internet
joy
preposition
between
below
from
regarding
interjection
whew
Ouch!
Gosh!
Yehey!
conjunction
unless
but
although
because
adverb
gently
always
quickly
lately
verb
sing
answer
protect
pronoun
he/ she
it
that / those
us / we
you/ me / them



the PAST SIMPLE
USE
SHORT PAST FINISHED ACTION
I gave the to Henry.
PAST HABIT
She often played with him.
ONE ACTION AFTER ANOTHER
He got up, paid the bill, and left.
ADVERBS
Last
week
year
April
yesterday
in 1980
3 months ago
...
BE
SINGULAR
I was
You were
He was
She
PLURAL
We were
You were
They were
+ FORM
(REGULAR) VERB+ED
playED
visitED
watchED
(IRREGULAR)
2nd
(Learn by heart!!)
Fly - flew - flown
Go - went - gone
- FORM
Subject
DIDN'T
VERB...
Jack didn't play...
I didn't go...
? FORM
(WH-)
DID
SUBJECT
VERB
?
(What) Did Jack play?
(Where) Did you go?
Photo by: Ana Todorovic@yahoo.com


علامات الرفع
الضمة
الاسم المفرد
جمع التكسير
جمع المؤنث السالم
الفعل المضارع الذي لم يتصل بآخره شيء
الواو
جمع المذكر السالم
الأسماء الخمسة
أبوك
أخوك
حموك
فوك
ذو مالٍ
الألف
في التثنية فقط
ثبوت النون
يفعلان
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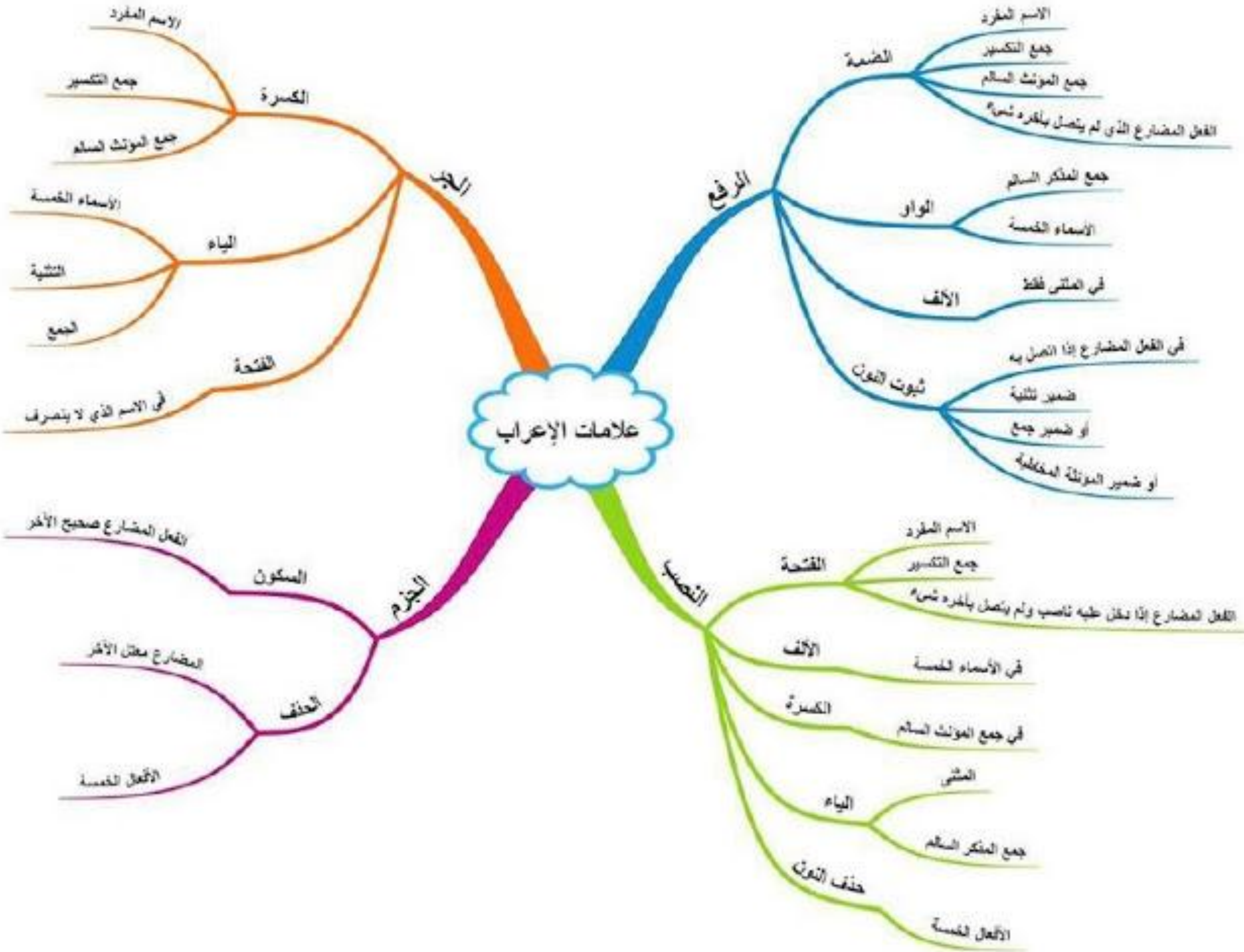
علامات الإعراب
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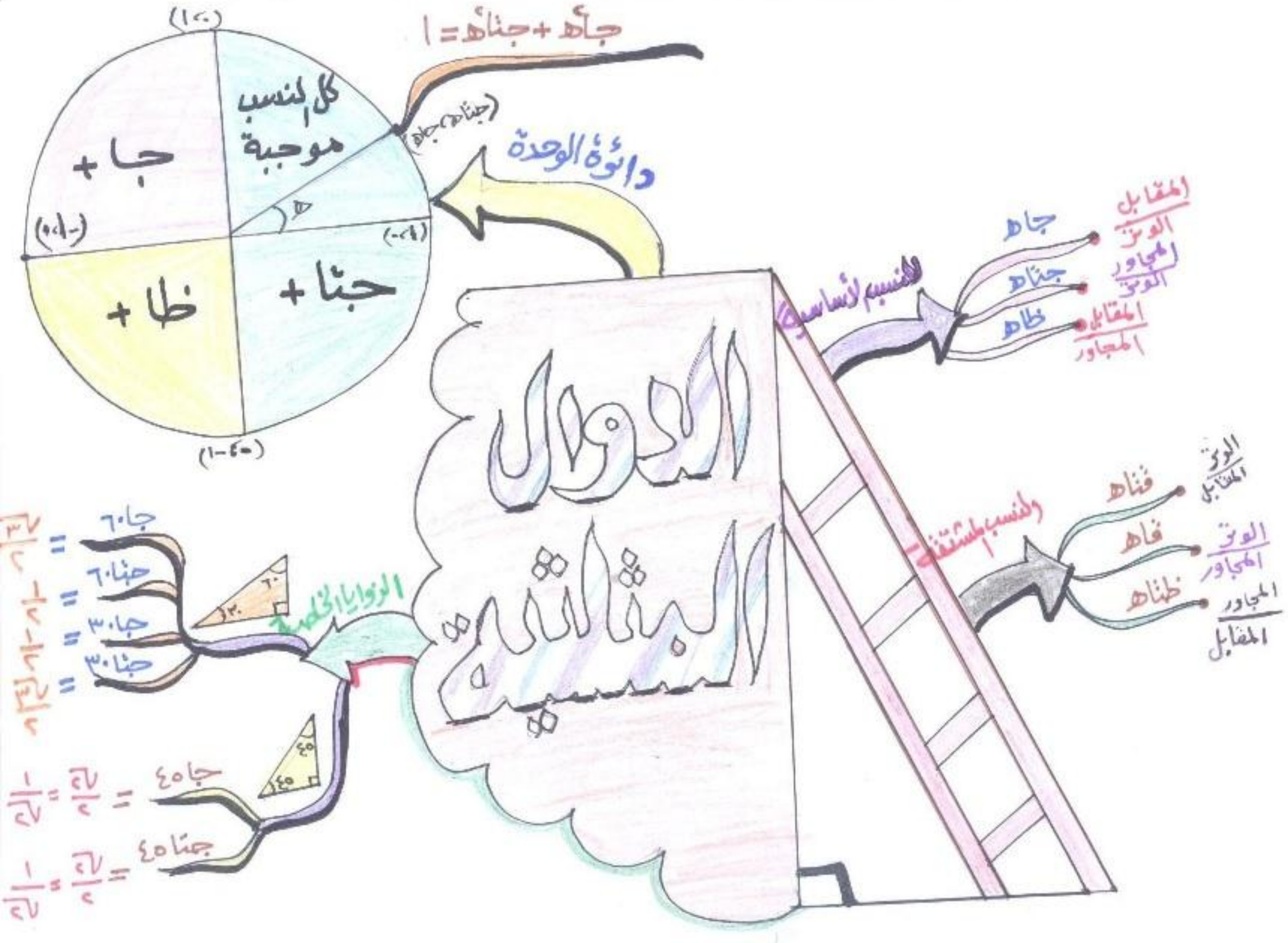
الواو
جمع المذكر السالم
الأسماء الخمسة
الألف
في المثنى فقط
ثبوت النون
في الفعل المضارع إذا اتصل به
ضمير تثنية
أو ضمير جمع
أو ضمير المؤنثة المخاطبة
النصب
الفتحة
الاسم المفرد
جمع التكسير
الفعل المضارع إذا دخل عليه ناصب ولم يتصل بآخره شيء
الألف
في الأسماء الخمسة
الكسرة
في جمع المؤنث السالم
الياء
المثنى
جمع المذكر السالم
حذف النون
الأفعال الخمسة
الجر
الكسرة
الاسم المفرد
جمع التكسير
جمع المؤنث السالم
الياء
الأسماء الخمسة
التثنية
الجمع
الفتحة
في الاسم الذي لا ينصرف
الجزم
السكون
الفعل المضارع صحيح الآخر
الحذف
المضارع معتل الآخر
الأفعال الخمسة



الدوال المثلثية
دائرة الوحدة
جا²هـ + جتا²هـ = ١
(جتاهـ، جاهـ)
كل النسب موجبة
جا +
ظا +
جتا +
هـ
النسب الأساسية
جاهـ
المقابل / الوتر
جتاهـ
المجاور / الوتر
ظاهـ
المقابل / المجاور
النسب المشتقة
قتاهـ
الوتر / المقابل
قاهـ
الوتر / المجاور
ظتاهـ
المجاور / المقابل
الزوايا الخاصة
جا٦٠
جتا٦٠
جا٣٠
جتا٣٠
جا٤٥
جتا٤٥
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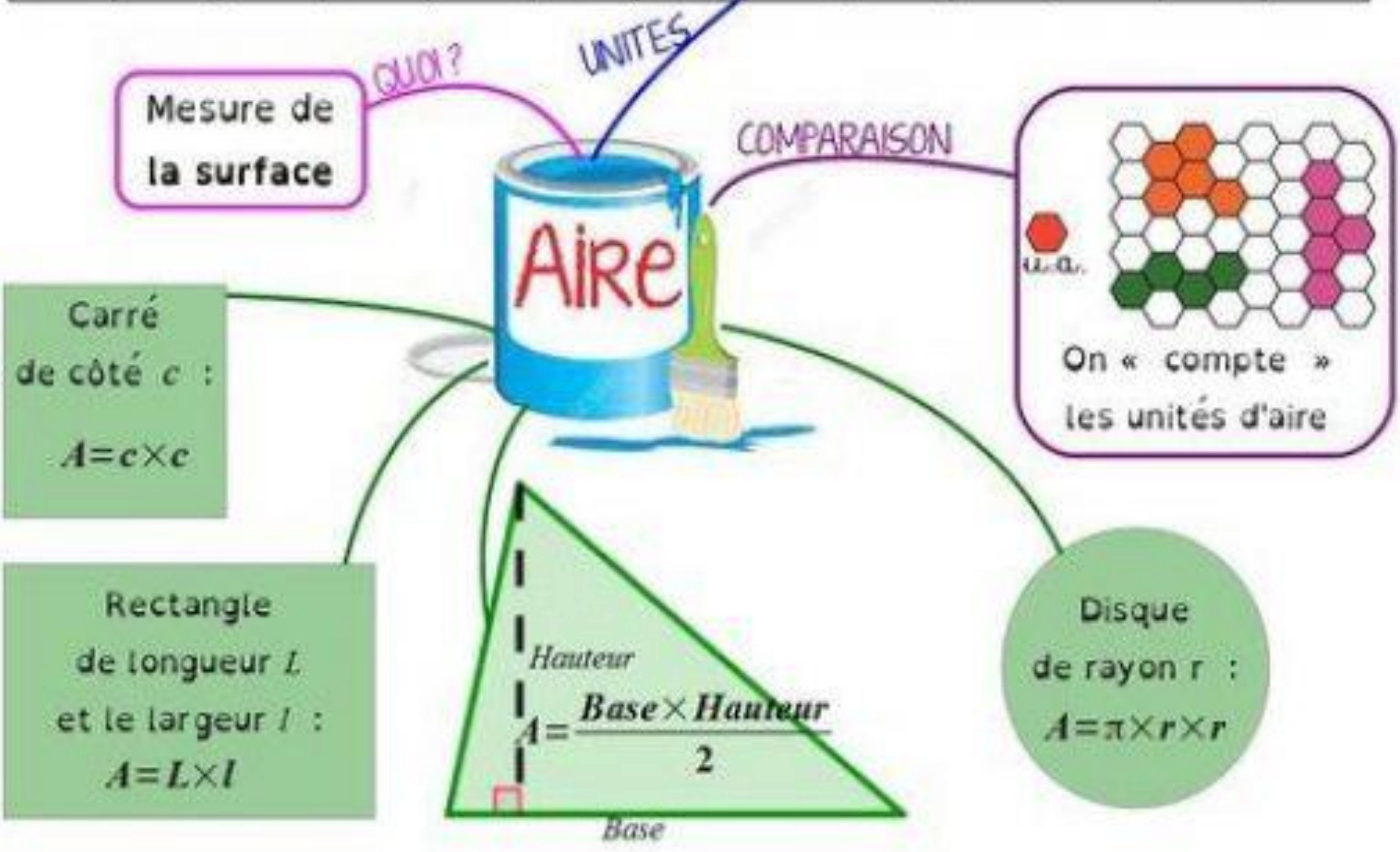
٣٠
٤٥

km²	hectare hm²	are dam²	centiare m²	dm²	cm²	mm²



9- Word Journal :

First, **summarize** the entire **topic on** paper with a **single word**.

Then use a **paragraph** to **explain** your **word** choice.

.

10- Opposites :

Instructor lists out **one or more concepts**,

for which **students** must come up with an **antonym**, and then **defend** their **choice**..

11-Four Corners :

be used as a formative assessment.

Students are asked a question.

In each of the four corners of the classroom, an opinion or response is posted.

Students express their response by standing in front of one of four statements, and then talking to others about why they have chosen their corner.

Four Corners promotes listening, verbal communication, critical thinking, and decision-making.

12-Think/Pair/Share (5-7 minutes) :

Pose a question or problem.
This should require participant to *explain a concept in their own words* or *evaluate what they've learned.*

Give participants **one minute** to **THINK** about their answers **individually**.

Have them **PAIR** with a **partner** to compare answers.

Ask them to **SHARE** their responses with the class

13- Picture Prompt (5-7 minutes):

Show students an **image** with no explanation, and ask them to identify/explain it, and justify their answers.

Or ask students to write about it using terms from lecture, or to name the processes and concepts shown.

Also works well as group activity. Do not give the "answer" until they have explored all options first..

14-KWL

- The KWL Chart starts students thinking about what they **Know** about a topic.

- what they **Want to know**.

- and what they have **Learned** in the end.

- This note-taking device guides students through a three-step process to activate background knowledge, develop a purpose for learning, and summarize

What I Know	What I Want to Know	What I Learned

15- Role Play :

There are two types of role-plays:

1- **controlled role-play**(for lower classes in which the dialogues, roles and situation is fixed).

2- **Free role-play**(for upper primary classes in which a situation is given and the dialogues, roles and characters are decided by the students).

16- 3-2-1

This technique provides a structure for students to record their own comprehension and summarize their learning.

It also gives teachers the opportunity to identify areas that need re-teaching, as well as areas of student interest.

- **1. Three** After the lesson, have each student record three things he/she **learned** from the lesson.
- **2. Two** Next, have students record two things that they found interesting and that they'd like to **learn more about.**
- **3. One** Then, have students record **one question they still have about it.**

17-Back and Forth

- This technique allows students to **explain** a **concept** or idea and **share** thoughts **with a partner.**
- Students not only explain their understanding of the concept, but are able to **listen to a classmate's explanation and check it for accuracy.**
- In this way, students are able to help each other correct **misunderstandings.**

18- Peer Review Writing Task:

To assist students with a **writing assignments**, encourage them to **exchange** drafts with a **partner**.

The partner reads the essay and writes a **three- paragraph response**: the first paragraph outlines the strengths of the essay, the second paragraph discusses the essay's problems, and the third paragraph is a description of what the partner would focus on in revision, if it were her essay.

19-Brain Drain :

Divide students into groups of(٥ gps). Hand out an empty grid of five rows and three columns to every student. Provide a task at the top to brainstorm. Each person brainstorms possible answers in row one.

After three minutes, rotate papers clockwise and work on row 2 (but do not repeat any answers from row 1).

Continue until sheet is filled in, then debrief to find the best answers.

20- Balloon Pop :

Give each group an inflated balloon with the task/problem trapped inside on a piece of paper. At the signal, all groups pop their balloons. Injects fun, noise, and energy to a group assignment..

٢١- mysterious:

The most beautiful experience we can have is the **mysterious**. It is the fundamental emotion that stands at the cradle of true art and true science.”

Albert Einstein


Dr DEREK MULLER
DUNLOP

- Who is working?
  - Instructor
- **Who is learning most?**
  - Instructor

What Works?

- If what you are doing is not working, CHANGE IT!!

- Doing the same thing over and over and expecting a different result is definition of INSANITY.
  - – Albert Einstein

Thank You